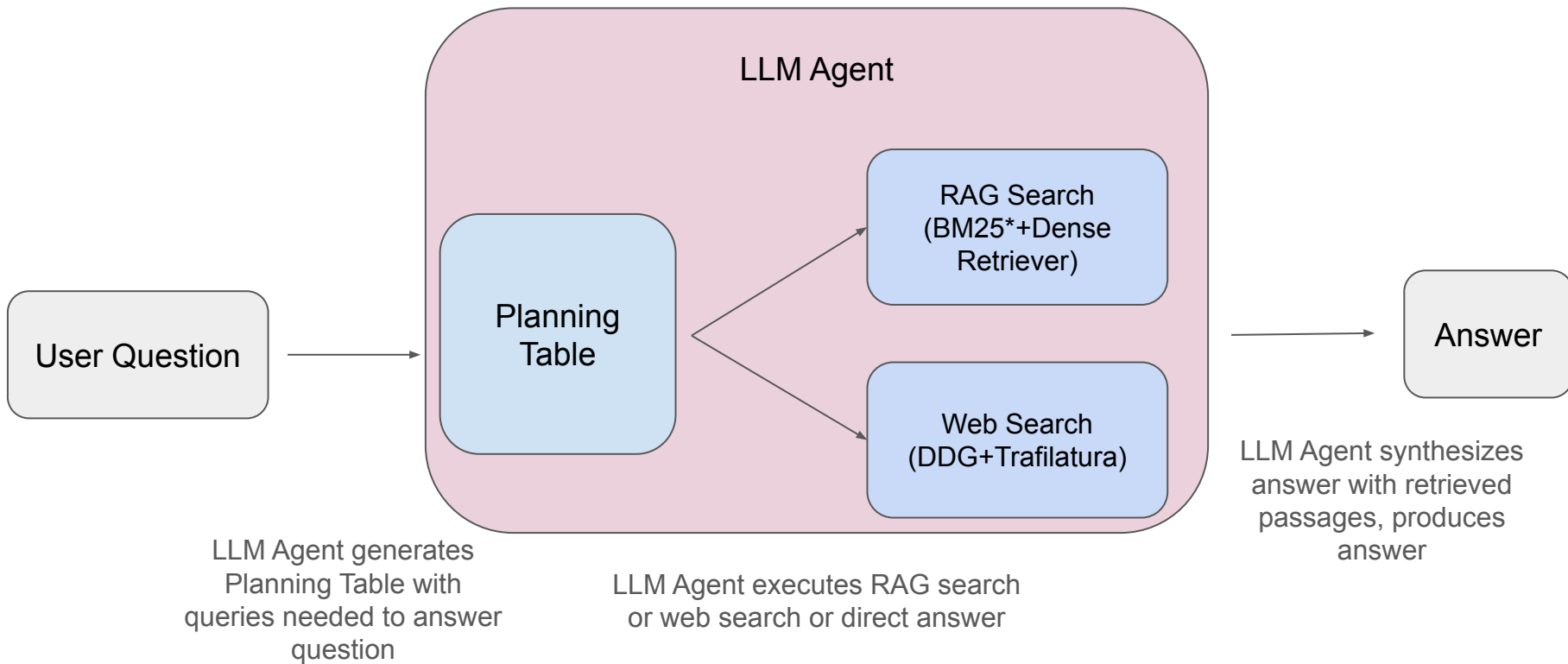


Agentic agent

Current status, progress, and issues

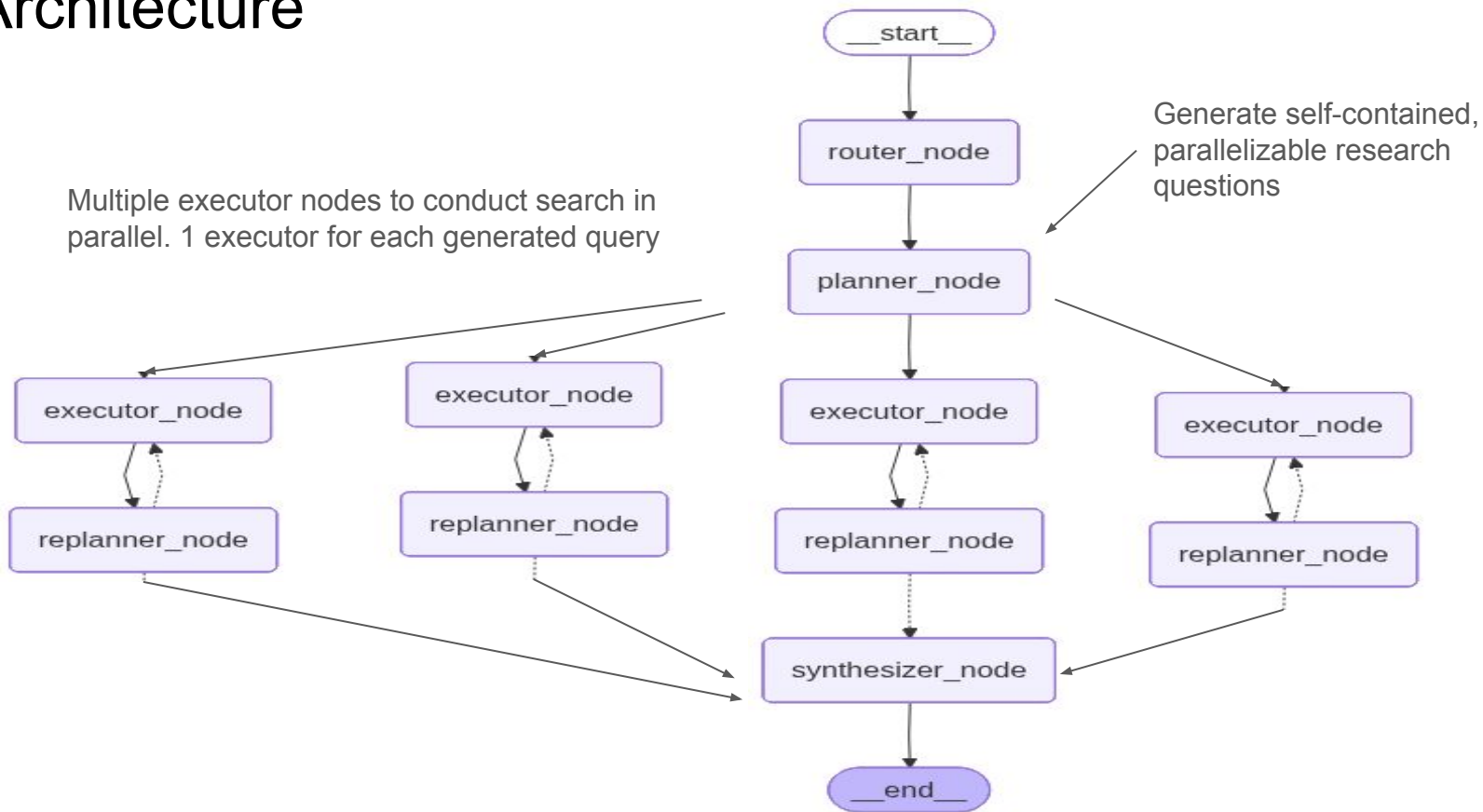
System Description



System Description

- **RAG Search:**
 - For each query (3 total), uses BM25* and gte-large-en-v1.5 retriever to retrieve 15 passages each → produces 45 candidate passages
 - Deduplicates 45 candidate passages, then uses ms-macro-MiniLM-L cross encoder to get top 5 passages from remaining candidate passages
- **Web Search:**
 - DuckDuckGo API to retrieve top 5 search results.
 - Contains brief description of source, not enough to answer question
 - Use trafilatura web scraper to scrape 2 most relevant URLs
- **Model used: deepseek-chat V3.2**
 - Encoder model: Alibaba-NLP/gte-large-en-v1.5
 - Cross encoder: cross-encoder/ms-marco-MiniLM-L-6-v2
- **Embedding DB**
 - Chromadb stores vectordb

Agent Architecture



Agent Architecture

- **Router Node:** LLM picks which ChromaDB collection to search — legal_passages (686k bar exam passages) or housing_statutes (1.8M housing statutes). Defaults to legal_passages.
- **Planner Node:** LLM decomposes the question into 1–5 self-contained research steps based on complexity. Each step gets a sub-question, action type (rag_search, web_search, or direct_answer), and retry budget (0–2).
- **Executor Node:** Runs all steps in parallel.
 - Per step: (1) retrieves — RAG rewrites into 3 queries, pools ~45 candidates, cross-encoder picks top 5; web searches DuckDuckGo + scrapes 2 URLs; direct skips retrieval.
 - (2) LLM synthesizes a cited sub-answer.
 - (3) LLM judges sufficiency — if insufficient, deterministic escalation within the step: rag → rewrite → rag → direct answer.
- After all steps finish, evidence is deduplicated and merged.
- **Synthesizer Node:** LLM combines all steps and evidence into an IRAC answer with citations. Then a separate LLM completeness check decides if more research is needed — if incomplete, routes to replanner; if complete or max rounds (3) hit, ends.
- **Replanner Node:** Programmatic (no LLM). Takes missing topics from the completeness check, builds a follow-up brief of covered/weak/gap areas, and routes back to planner for targeted new steps.

Agent Skills Used by Nodes

Planner: Decomposes the question into 1–5 research steps, assesses complexity, chooses search method (rag_search, direct_answer, or web_search), and sets retry budgets.

Query-rewrite: Rewrites one sub-question into 1 primary + 2 alternate dense keyword queries using legal terminology (MBE, doctrinal labels, retrieval hints).

Synthesize-and-cite: Produces a 100–250 word cited sub-answer per step, distinguishing passage-backed claims ([Source N]) from doctrine-based reasoning.

Judge: Evaluates if retrieved passages are sufficient, partial, or insufficient. Controls per-step escalation; suggests rewritten query if insufficient.

Verifier: Same role as judge for direct_answer steps. Evaluates if LLM's doctrinal reasoning is grounded and appropriately hedged.

Synthesizer: Combines all steps and evidence index into a final IRAC answer (Issue/Rule/Application/Conclusion) with [Evidence N] citations.

Benchmark overview

- Bar Exam QA: historical Multistate Bar Exam multi-choice questions with gold passages to support answers
 - Gold passages curated by law students (explanation passage unavailable with historical MBEs)
- 850k Passage Pool including:
 - Curated gold passages
 - 2019-2021 US caselaws
 - Wex
 - Cornell Law School

Bar Exam QA Dataset Sample

idx	dataset	example_id	prompt_id	source	subject	question_number
mbe_4	mbe	4	2	MBE-1972-78-part2		5

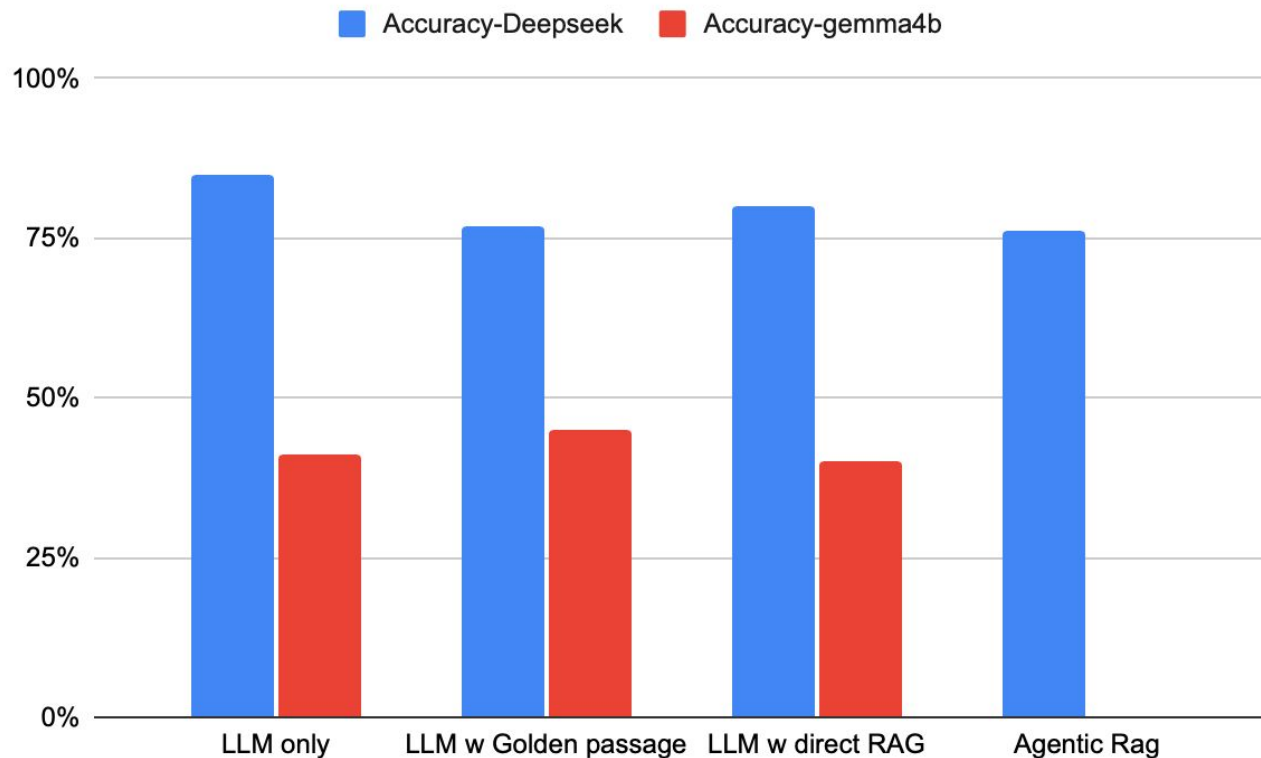
prompt	question	choice_a	choice_b	choice_c	choice_d	answer	gold_passage	gold_idx
Sue owned a five-acre tract of land, one acre of which had previously been owned by Opal, but to which Sue had acquired title by adverse possession. Sue contracted to convey the full five-acre tract to Peg, but the contract did not specify the quality of title Sue would convey. At closing, Peg refused the tendered deed and demanded return of her earnest money.	Suppose Peg pays the purchase price and accepts a deed. Subsequently, Sue's title to the one acre proves inadequate and Opal ejects Peg from that acre. Peg sues Sue for damages. Which of the following statements applies most accurately to the determination of Peg's rights?	Sue's deed was fraudulent.	The terms of the deed control Sue's liability.	The only remedy available for breach of warranty of title is rescission.	Peg's rights are based on the implied covenant that the title conveyed shall be marketable.	B	They are distinguishable from Schmidt & Wilson, 100 So.2d 123 (Fla. 1st DCA 1962).	mbe_2318

Benchmarks: Setup

100 question eval on BarExam-qa dataset, measure performance on no-RAG, simple-RAG, agentic-RAG using DeepSeek model

- **LLM Only**
 - Provide the LLM only with the question and multiple choice
- **LLM with Golden Passage**
 - Provide the LLM with the question, multichoice, and golden passage
- **Direct RAG**
 - LLM with vanilla RAG + query rewriting (primary query + 2 rewrites)
 - 2 alternative query are rewritten from primary query but using synonyms, related doctrines, alternative legal terms
 - Rag database: contains all embedded passages from BarExam dataset
- **Agentic RAG (ours)**
 - 5-node LangGraph pipeline, planner decomposition, parallel execution, per-step judge+escalation, completeness loop, IRAC synthesis.

Benchmarks Performance (with DeepSeek v3.2 and Gemma3-4b)



LLM: deepseek-chat

100-q Eval Accuracies:

Deepseek:

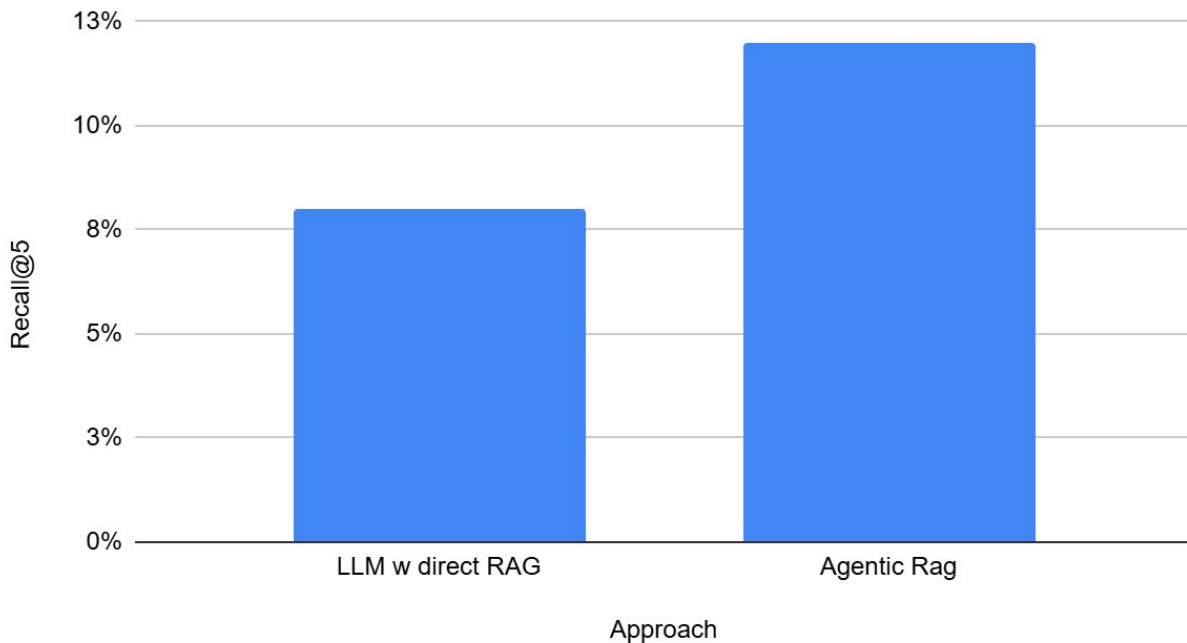
- LLM-only: 85%
- LLM with golden passage: 77%
- LLM with direct RAG: 80%
- Agentic RAG: 76%

Gemma3-4b:

- LLM-only: 41%
- LLM with golden passage: 45%
- LLM with direct RAG: 40%
- Agentic RAG: In progress

Benchmarks Performance - Golden Passage Retrieval

Direct vs Agentic RAG Recall@5 Performance



LLM: deepseek-chat

Recall@5 for gold
passage:

DeepSeek:

- LLM with direct RAG: 8%
- Agentic RAG: 15%

Gemma3-4b

- LLM with direct RAG: 2%
- Agentic RAG: In progress

Synthesizer Skill Snippet

Task

Produce a cohesive, IRAC-structured legal answer that:

1. Directly addresses the original question
2. Grounds every claim in the provided evidence (no fabrication)
3. Maps citations to specific evidence entries using `[Evidence N]` format

Critical Rules

1. **Evidence-first, knowledge-supplemented.** Ground your answer primarily in the research findings and evidence index. Cite evidence as `[Evidence N]`. If the evidence is incomplete, you MAY supplement with established legal doctrine — but clearly distinguish: use `[Evidence N]` for passage-derived claims, and phrases like "Under established doctrine" for knowledge-based claims. Never fabricate specific statutes, dollar amounts, dates, or case names.
2. **Synthesize across steps.** Weave together findings from multiple research steps into a unified narrative. Do not just concatenate step results — integrate them coherently.
3. **Step attribution when useful.** When different steps addressed different aspects, briefly attribute: "Regarding the duty element [Evidence 1][Evidence 2]...; on causation [Evidence 4]..."
4. **Multiple-choice selection.** If the original question contains "Answer choices:", end your response with a blank line followed by: **Answer: (X)** selecting the letter best supported by your analysis. Use both the evidence and your legal knowledge to reason through the answer choices.

Judge Skill Snippet

Retrieval Sufficiency Judge

You are evaluating whether a set of retrieved passages (from a legal corpus or web search) is sufficient to support the sub-question's contribution to the original research goal.

Evaluation Criteria

Evaluate along four dimensions:

1. **Relevance**: Do the passages address the sub-question's legal doctrine, element, or rule? Passages that only tangentially relate to the topic (e.g., discuss a different doctrine using the same word) are not sufficient.
2. **Abstraction level**: Are the passages at the doctrine level (general legal rules, elements, definitions) rather than hyper-specific case facts that may not generalize? Doctrine-level passages are more useful for legal synthesis.
3. **Jurisdiction fit**: If the question is jurisdiction-specific, do the passages address the right jurisdiction? If jurisdiction is not specified, general common-law passages are acceptable.
4. **Completeness**: Do the passages, taken together, give enough information to make a grounded claim in the final answer? A partial answer is acceptable if combined with other completed steps — assess only whether this step contributed meaningfully.
5. **Direct support vs analogy**: If the passages discuss only adjacent doctrines, related concepts, or loose analogies, that is NOT sufficient. The retrieval must support the actual targeted doctrine, rule, element, current fact, or legal standard.

Output Format

Return ONLY valid JSON — no prose, no markdown fences:

```
```json
{
 "sufficient": "full",
 "reason": "One sentence explaining the verdict.",
 "missing": null,
 "suggested_rewrite": null
}
```
```

- `sufficient`: One of three values:

- `"full"` — the passages comprehensively address the sub-question.
 - `"partial"` — the passages address some aspects of the sub-question but are missing specific components. The evidence found IS useful and should be kept, but there are identifiable gaps.
 - `false` — the passages miss the key doctrine entirely or are too tangential to be useful.
- `reason`: Brief explanation tied to the specific sub-question and passages. Do not reference confidence scores.
- `missing`: If `sufficient` is `"partial"`, briefly state what specific information is missing (e.g., "missing the standard for breach" or "covers the rule but not the exceptions"). Set to `null` for `"full"` or `false`.
- `suggested_rewrite`: If `sufficient` is `false` **and** `action_type` is `rag_search`, provide an alternative query phrasing. Set to `null` otherwise.

Replanner Skill Snippet

Legal Research Replanner

You are deciding whether to continue, complete, or retry a research step, given that the judge has already determined the last step's retrieval was **sufficient**.

Your Decision

Choose one of three actions:

`next`

The last step produced useful evidence. Proceed to the next pending step.

Use when:

- The result substantively addresses the sub-question
- The accumulated evidence is growing toward a complete answer
- Pending steps are still needed to address aspects of the original question

`complete`

All necessary research has been gathered. Route to final synthesis.

Use when:

- All pending steps are done AND accumulated evidence can answer the original question
- Accumulated evidence already answers the original question (no need to execute remaining pending steps)
- Research has stalled despite escalation (judge has consistently found results insufficient across action types)

`retry`

Use this when the judge marked the last step sufficient, but you can see that the research direction is still wrong for the original question and a different phrasing would better serve the overall goal.

Only use `retry` when:

- The result is substantively off-topic relative to the original question (not just the sub-question)
- A more abstract or differently-worded query would clearly serve the research goal better

Retry limit enforced in code: If the step has already been retried once, the code will override `retry` with `next` or `complete`. Do not retry a step more than once.

When retrying, you MUST provide `revised\_question` — a rephrased version of the sub-question. Make it:

- More abstract (move from fact-specific → doctrine-level)
- Different terminology (if "exclusionary rule" failed, try "suppression of evidence Fourth Amendment")

Future steps/ongoing issues